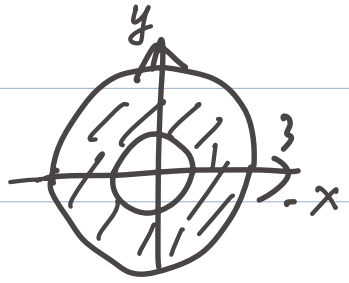
§ 9.1 多元函数.

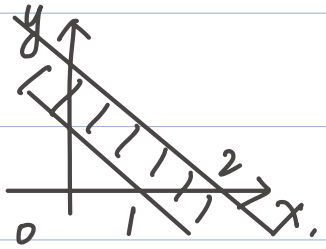
[P158. Q2].



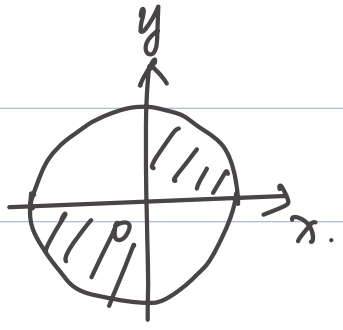
(2) $D=\{(x,y)\mid 1\le x^2+y^2\le 9\}$.

(4) $-1\le\sqrt{2-x-y}\le 1$

$D=\{(x,y)\mid 1\le x+y\le 2\}$.



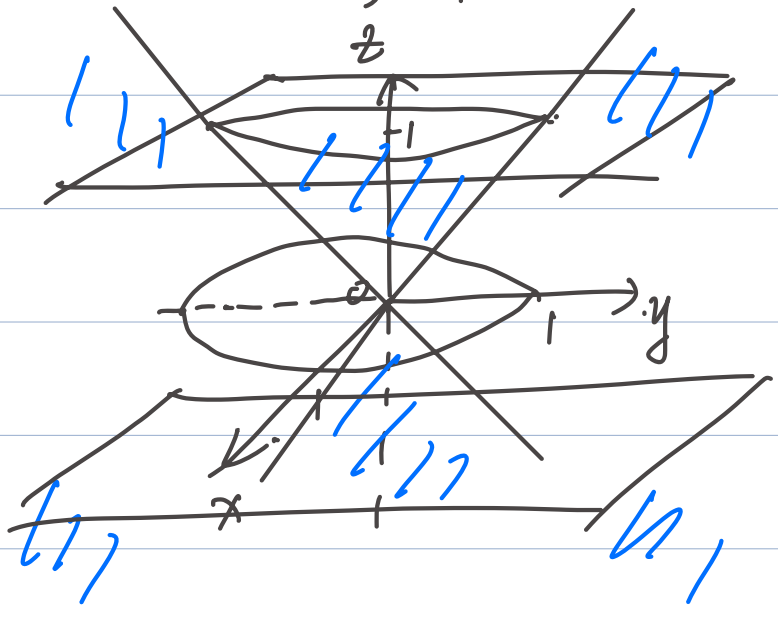
(6) $\begin{cases}\frac{y}{x}>0\\ x^2+y^2\le 1\end{cases}$



□

[P158. Q3]

(3) $D=\{(x,y,z)\mid \sqrt{x^2+y^2}<|z|<1$ 或 $1<|z|<\sqrt{x^2+y^2}\}$.



[P158. Q4].

$f(x+\frac{1}{x},y-1)=x^2+2xy+y^2+\frac{1}{x^2}+\frac{2y}{x}-2(x+y)-\frac{2}{x}+4$

$=x^2+\frac{1}{x^2}+2y\left(x+\frac{1}{x}\right)+y^2-2y+4-2\left(x+\frac{1}{x}\right)$

$$\Rightarrow f(x,y) = x^2-2 + 2(y+1)x + y^2+3-2x$$

$$= x^2+2xy+y^2+1. \qquad \square$$

§9.2 二重极限.

[P159. Q6]

(2) $$\left|\frac{2x+y+2}{2-x-2y}-1\right| = \frac{3|x+y|}{|2-x-2y|}$$

为使 $\frac{3|x+y|}{|2-x-2y|}<\varepsilon$, 可取 $\begin{cases} |x+y|<\frac{\varepsilon}{3} \\ |2-x-2y|>1 \end{cases} \Leftarrow \begin{cases} |x|<\frac{\varepsilon}{6} \\ |y|<\frac{\varepsilon}{6} \\ |x+2y|<1 \end{cases}$

故对给定的 $\varepsilon>0$, 令 $\delta = \min\left\{\frac{\varepsilon}{6}, \frac{1}{3}\right\}$.

当 $0<\sqrt{x^2+y^2}<\delta$ 时, $|x|<\delta$, 于是 $|x+y|<2\delta<\frac{\varepsilon}{3}$

$|y|<\delta$ $|x+2y|<1$.

$|2-x-2y|>1$.

从而 $\left|\frac{2x+y+2}{2-x-2y}-1\right|<\varepsilon$.

故由二元极限的定义, $\lim\limits_{(x,y)\to(0,0)} \frac{2x+y+2}{2-x-2y} = 1$. $\square$

[P159. Q7]

(1) $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{x+y}{|x|+|y|}$ 不存在.

① $y=x>0$. $\dfrac{x+y}{|x|+|y|}=1$.

② $y=x<0$. $\dfrac{x+y}{|x|+|y|}=-1$

(4) $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{1-\cos(xy)}{x^2+y^2} = \lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{\frac{1}{2}x^2y^2}{x^2+y^2}$ (等价替换).

由于 $0\le \dfrac{\frac{1}{2}x^2y^2}{x^2+y^2} \le \dfrac{\frac{1}{2}x^2y^2}{2|x||y|} = \dfrac{1}{4}|x||y| \to 0$.

故由夹逼定理, $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{1-\cos(xy)}{x^2+y^2}=0$.

(7) $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{x^2y^2}{x^2y^2+(x-y)^2} = \lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{1}{1+\left(\frac{1}{y}-\frac{1}{x}\right)^2}$.

① $y=x$. $\dfrac{1}{1+\left(\frac{1}{y}-\frac{1}{x}\right)^2}\to 1$

② $y=-x$. $\dfrac{1}{1+\left(\frac{1}{y}-\frac{1}{x}\right)^2} = \dfrac{1}{1+\frac{4}{x^2}} \to 0$.

极 $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{x^2y^2}{x^2y^2+(x-y)^2}$ 不存在.

(8) $0\le \dfrac{x^4+y^4}{x^2+y^2} \le \dfrac{2\left(\max\{|x|,|y|\}\right)^4}{\left(\max\{|x|,|y|\}\right)^2} = 2\left(\max\{|x|,|y|\}\right)^2 \to 0.$

故由夹逼定理, $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{x^4+y^4}{x^2+y^2} = 0.$ □

[P159. Q10].

$$\left|\frac{xy^2}{x^2+y^2}\right| \le \frac{|x|y^2}{2|x||y|} = \frac{1}{2}|y| \to 0.$$

$\Rightarrow \lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{xy^2}{x^2+y^2} = 0 \Rightarrow$ 连续. □

§9.3. 偏导. 全微分.

[P159. Q12].

(1) $f(x,y) = x + (y-1)\arctan\dfrac{x}{y}.$

$$f_x'(0,1) = \lim_{\Delta x\to 0} \frac{f(\Delta x,1)-f(0,1)}{\Delta x}$$

$$= \lim_{\Delta x\to 0} \frac{\Delta x - 0}{\Delta x} = 1.$$

$$f'_y(0,1)=\lim_{\Delta y\to 0}\frac{f(0,1+\Delta y)-f(0,1)}{\Delta y}$$

$$=\lim_{\Delta y\to 0}\frac{0-0}{\Delta y}=0.$$ □

[P15, Q13]

(2) <方法一：偏导>

$$z=e^{\frac{y}{x}}(x+y).$$

$$\frac{\partial z}{\partial x}=\frac{\partial}{\partial x}e^{\frac{y}{x}}\cdot(x+y)+e^{\frac{y}{x}}\cdot 1$$

$$=e^{\frac{y}{x}}\left(-\frac{y}{x^2}\right)(x+y)+e^{\frac{y}{x}}=e^{\frac{y}{x}}\left(-\frac{y}{x}-\frac{y^2}{x^2}+1\right)$$

$$\frac{\partial z}{\partial y}=e^{\frac{y}{x}}\cdot\frac{1}{x}(x+y)+e^{\frac{y}{x}}\cdot 1=e^{\frac{y}{x}}\left(2+\frac{y}{x}\right).$$

<方法二：全微分>.

$$dz=de^{\frac{y}{x}}\cdot(x+y)+e^{\frac{y}{x}}\,d(x+y)$$

$$=e^{\frac{y}{x}}\,d\left(\frac{y}{x}\right)\cdot(x+y)+e^{\frac{y}{x}}(dx+dy)$$

$$=e^{\frac{y}{x}}\cdot\frac{x\,dy-y\,dx}{x^2}(x+y)+e^{\frac{y}{x}}(dx+dy)$$

$$=e^{\frac{y}{x}}\left[\left(-\frac{y}{x}-\frac{y^2}{x^2}+1\right)dx+\left(2+\frac{y}{x}\right)dy\right]$$

$$\Rightarrow z_x=e^{\frac{y}{x}}\left(-\frac{y}{x}-\frac{y^2}{x^2}+1\right).$$

$$z_y = e^{\frac{y}{x}}\left(2+\frac{y}{x}\right).$$

(3) $$dz = \frac{2dx + \frac{x\,dx + y\,dy}{\sqrt{x^2+y^2}}}{2x+\sqrt{x^2+y^2}} \Rightarrow \begin{cases} z_x = \dfrac{2+\frac{x}{\sqrt{x^2+y^2}}}{2x+\sqrt{x^2+y^2}} \\ z_y = \dfrac{y}{\sqrt{x^2+y^2}\left(2x+\sqrt{x^2+y^2}\right)} \end{cases}.$$

(6) $$u_x = -\frac{x}{(x^2+y^2+z^2)^{3/2}}$$

$$u_y = -\frac{y}{(x^2+y^2+z^2)^{3/2}}$$

$$u_z = -\frac{z}{(x^2+y^2+z^2)^{3/2}}.$$

□

[P160. Q15]

$$f_x'(1,0) = \lim_{\Delta x\to 0}\frac{f(1+\Delta x,0)-f(1,0)}{\Delta x} = \lim_{\Delta x\to 0}\frac{\Delta x\cdot 100!\cdot(-99!)}{\Delta x}$$

$$= -100!\cdot 99!$$

$$f_{xy}''(1,1) = \lim_{\Delta y\to 0}\frac{f_x'(1,1+\Delta y)-f_x'(1,1)}{\Delta y}$$

$$f_x'(1,1+\Delta y) = \lim_{\Delta x\to 0}\frac{f(1+\Delta x,1+\Delta y)-f(1,1+\Delta y)}{\Delta x}$$

$$\frac{\Delta x\cdot\Delta y\cdot(-99!)\cdot(-99!)}{\Delta x\cdot\Delta y}$$

$$= \lim_{\Delta x \to 0} \frac{\Delta x \cdot \Delta y \cdot \cdots \cdot (-1)!}{\Delta x}$$

$$= (99!)^2 \Delta y.$$

$$\Rightarrow f''_{xy}(1,1) = (99!)^2. \qquad \square$$

[P160. 习7]

$$f'_x(0,0) = \lim_{\Delta x \to 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x} = \lim_{\Delta x \to 0} \frac{(\Delta x)^2 \sin \frac{1}{|\Delta x|}}{\Delta x} = \lim_{\Delta x \to 0} \Delta x \sin \frac{1}{|\Delta x|}$$

$$= 0$$

(无穷小量乘有界量).

当 $(x,y) \neq (0,0)$ 时,

$$f'_x(x,y) = 2x \sin \frac{1}{\sqrt{x^2+y^2}} + (x^2+y^2) \cdot \cos \frac{1}{\sqrt{x^2+y^2}} \cdot \frac{-x}{(x^2+y^2)^{3/2}}.$$

$$= 2x \cdot \sin \frac{1}{\sqrt{x^2+y^2}} - \frac{x}{\sqrt{x^2+y^2}} \cdot \cos \frac{1}{\sqrt{x^2+y^2}}.$$

无穷小乘有界量　　?　　$\to 1$

当 $y=0$ 且 $x \to 0^+$ 时, $f_x'(x,y) \to 0 - 1 \cdot 1 = -1$

当 $y=0$ 且 $x \to 0^-$ 时, $f_x'(x,y) \to 0 - (-1) \cdot 1 = 1$.

故 $\lim\limits_{\substack{x \to 0 \\ y \to 0}} f'_x(x,y)$ 不存在. 因此 $f_x'(x,y)$ 在 $(0,0)$ 处不连续. $\square$

[P160, Q19].

$z = f(x^2 - y^2)$.

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}\left(f'(x^2-y^2)\cdot 2x\right)$$

$$= -4xy f''(x^2-y^2).$$ □

[P160, Q20].

$z = xy + xf\left(\frac{y}{x}\right)$.

$$dz = x\,dy + y\,dx + \underbrace{x\,df\left(\frac{y}{x}\right)}_{\parallel} + f\left(\frac{y}{x}\right)dx$$

$$xf'\,d\frac{y}{x} = xf'\,\frac{x\,dy - y\,dx}{x^2}$$

$$\Rightarrow z_x = y - \frac{y}{x}f' + f.$$

$$z_y = x + f'.$$

$$\Rightarrow xz_x + yz_y = xy - yf' + xf + xy + yf'$$

$$= xy + (xy + xf) = xy + z.$$ □

[P160, Q23].

$z = e^{-\left(\frac{1}{x} + \frac{1}{y}\right)}$.

$$z_x = e^{-(\frac{1}{x}+\frac{1}{y})} \cdot \frac{1}{x^2}$$
$$z_y = e^{-(\frac{1}{x}+\frac{1}{y})} \cdot \frac{1}{y^2}$$
$$\Rightarrow x^2 z_x + y^2 z_y = 2z.$$

□

§9.4 复合函数求导.

[P161. Q31]

$$\begin{cases} z = \sqrt{u^2+v^2} \\ u = x\sin y \\ v = e^{xy} \end{cases}$$

(1) <方法一：偏导>.

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$= \frac{u}{\sqrt{u^2+v^2}}\cdot \sin y + \frac{v}{\sqrt{u^2+v^2}} e^{xy}\cdot y$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}$$

$$= \frac{u}{\sqrt{u^2+v^2}}\cdot x\cos y + \frac{v}{\sqrt{u^2+v^2}}\cdot e^{xy}\cdot x$$

<方法二：全微分>.

$$dz = \frac{1}{\sqrt{u^2+v^2}}(u\,du + v\,dv)$$

$$= \frac{1}{\sqrt{u^2+v^2}}\left[u(x\cos y\,dy + \sin y\,dx) + v\cdot e^{xy}(x\,dy + y\,dx)\right]$$

$\Rightarrow \cdots$

(2) $z = e^{uv}$. $u = \frac{1}{2}\ln(x^2+y^2)$. $v = \arctan\frac{y}{x}$.

$$dz = e^{uv}(u\,dv + v\,du)$$

$$= e^{uv}\left(u\cdot\frac{1}{1+(\frac{y}{x})^2}\,d\frac{y}{x} + v\cdot\frac{x\,dx+y\,dy}{x^2+y^2}\right)$$

$$= e^{uv}\left(u\cdot\frac{x^2}{x^2+y^2}\cdot\frac{x\,dy-y\,dx}{x^2} + v\cdot\frac{x\,dx+y\,dy}{x^2+y^2}\right)$$

$$\Rightarrow z_x = \frac{e^{uv}}{x^2+y^2}(-uy+vx)$$

$$z_y = \frac{e^{uv}}{x^2+y^2}(ux+vy).$$

(3) $z = \ln(x^2+y^2)$. $x = t\cos t$. $y = -\sin t$.

$$dz = \frac{1}{x^2+y^2}(2x\,dx + 2y\,dy)$$

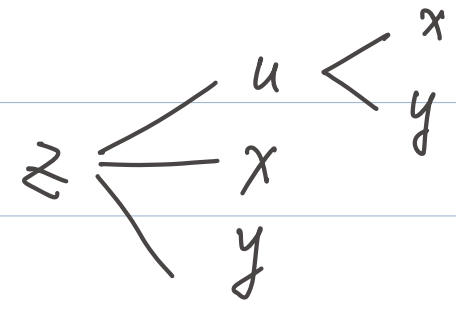
$$= \frac{2}{x^2+y^2}\left[x(-t\sin t\,dt + \cos t\,dt) - y\cos t\,dt\right]$$

$$\Rightarrow \frac{dz}{dt} = \frac{2}{x^2+y^2}\left[x(\cos t - t\sin t) - y\cos t\right].$$

□

[P161. Q33]

$z = f(u, x, y)$. $u = xe^y$.



$$\frac{\partial z}{\partial x} = f_1' \cdot \frac{\partial u}{\partial x} + f_2' = f_1' \cdot e^y + f_2'$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial f_1'}{\partial y} \cdot e^y + f_1' \cdot e^y + \frac{\partial f_2'}{\partial y}$$

$$= \left(f_{11}'' \frac{\partial u}{\partial y} + f_{13}''\right)e^y + f_1' e^y + \left(f_{21}'' \frac{\partial u}{\partial y} + f_{23}''\right)$$

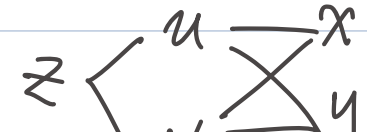
$$= e^y\left(f_{11}'' \cdot xe^y + f_{13}'' + f_1'\right) + f_{21}'' \cdot xe^y + f_{23}''.$$

□

[P161. Q34]

$z = f(u, v)$. $u = x^2 - y^2$. $v = x\sin y$.

$$\frac{\partial z}{\partial x} = f_1' \cdot \frac{\partial u}{\partial x} + f_2' \cdot \frac{\partial v}{\partial x}$$



$= 2xf_1' + \sin y \cdot f_2'$.

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = 2x \cdot \frac{\partial f_1'}{\partial y} + \cos y \cdot f_2' + \sin y \cdot \frac{\partial f_2'}{\partial y}$$

$$= 2x\left(f_{11}'' \frac{\partial u}{\partial y} + f_{12}'' \frac{\partial v}{\partial y}\right) + \cos y \cdot f_2' + \sin y \left(f_{21}'' \frac{\partial u}{\partial y} + f_{22}'' \frac{\partial v}{\partial y}\right)$$

$$= 2x\left(-2y f_{11}'' + x\cos y f_{12}''\right) + \cos y \cdot f_2' + \sin y \left(-2y f_{21}'' + x\cos y f_{22}''\right).$$

□

[P162. Q39].

<方法一：偏导>

① $$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}\right)$$

$$= \frac{\partial}{\partial x}\left(z_u + z_v\right) = \frac{\partial z_u}{\partial x} + \frac{\partial z_v}{\partial x}$$

$$= z_{uu} + z_{uv} + z_{vu} + z_{vv}$$

$$= z_{uu} + 2z_{uv} + z_{vv}.$$

② $$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y}\left(z_u + z_v\right) = z_{uu} \cdot 2 + z_{uv} \cdot a + z_{vu} \cdot 2 + z_{vv} \cdot a$$

$$= 2z_{uu} + (a+2)z_{uv} + a z_{vv}.$$

(3) $$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}\right)$$

$$= \frac{\partial}{\partial y}(2z_u + a z_v)$$

$$= 2z_{uu}\cdot 2 + 2z_{uv}\cdot a + a z_{vu}\cdot 2 + a z_v \cdot a$$

$$= 4z_{uu} + 4a z_{uv} + a^2 z_{vv}.$$

$$\Rightarrow 2z_{xx} + z_{xy} - z_{yy} = (6-3a)z_{uv} + (2+a-a^2)z_v = 0$$

因此 $\begin{cases} 6-3a \neq 0. \\ 2+a-a^2 = 0. \end{cases}$ 解得 $a = -1$.

(a=2 时 u,v 线性相关，退化为单自变量代换)

<方法二：线性算子>.

$$\begin{cases} \frac{\partial}{\partial x} = \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \\ \frac{\partial}{\partial y} = 2\frac{\partial}{\partial u} + a\frac{\partial}{\partial v} \end{cases}$$

简记成 $\begin{cases} \partial_x = \partial_u + \partial_v \\ \partial_y = 2\partial_u + a\partial_v \end{cases}$

于是 $2\partial_x^2+\partial_x\partial_y-\partial_y^2=(\partial_x\ \partial_y)\begin{pmatrix}2&1\\0&-1\end{pmatrix}\begin{pmatrix}\partial_x\\\partial_y\end{pmatrix}$

$=(\partial_u\ \partial_v)\begin{pmatrix}1&2\\1&a\end{pmatrix}\begin{pmatrix}2&1\\0&-1\end{pmatrix}\begin{pmatrix}1&1\\2&a\end{pmatrix}\begin{pmatrix}\partial_u\\\partial_v\end{pmatrix}$

$=(\partial_u\ \partial_v)\begin{pmatrix}0&2-a\\4-2a&a+2-a^2\end{pmatrix}\begin{pmatrix}\partial_u\\\partial_v\end{pmatrix}$

$\Rightarrow\cdots$ □

[P162.Q40].

$\varphi(x)=f(x,f(x,x))$

$\Rightarrow \frac{d}{dx}\varphi^3(x)=3\varphi^2(x)\cdot\varphi'(x)$

$=3\varphi^2(x)\left[f_1'+f_2'\cdot\frac{d}{dx}f(x,x)\right]$

$=3\varphi^2(x)\left[f_1'+f_2'(f_1'+f_2')\right]$

$=3\cdot1\cdot[2+3\cdot(2+3)]$

$=3\times17=51.$ □

(邪修方法: 取 $f(x,y)=2x+3y-4$

$f(x,x)=5x-4.$

$\varphi(x)=17x-16.$)